UNIVERSITY OF AZAD JAMMU & KASHMIR CHELLAH CAMPUS MUZAFFERABAD



DEPARTMENT OF SOFTWARE ENGINEERING

Digital Image Processing

(Lab Task # 01)

Submitted By:

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(2024-SE-37)

Submitted To:

Respected “Engr. Fahad Nisar”

Session: 2024-2028

Lab Task # 01 – Python Setup, Fundamentals, and Imports

Objective

To install Python and VS Code, verify the environment, install the introductory packages (NumPy, Pillow, Matplotlib), and verify the required import statements.

Section 1: Python & Pip Environment Verification

Verified the installation of the Python 3.15 runtime environment and the pip package manager on the Windows operating system using command prompt utilities.

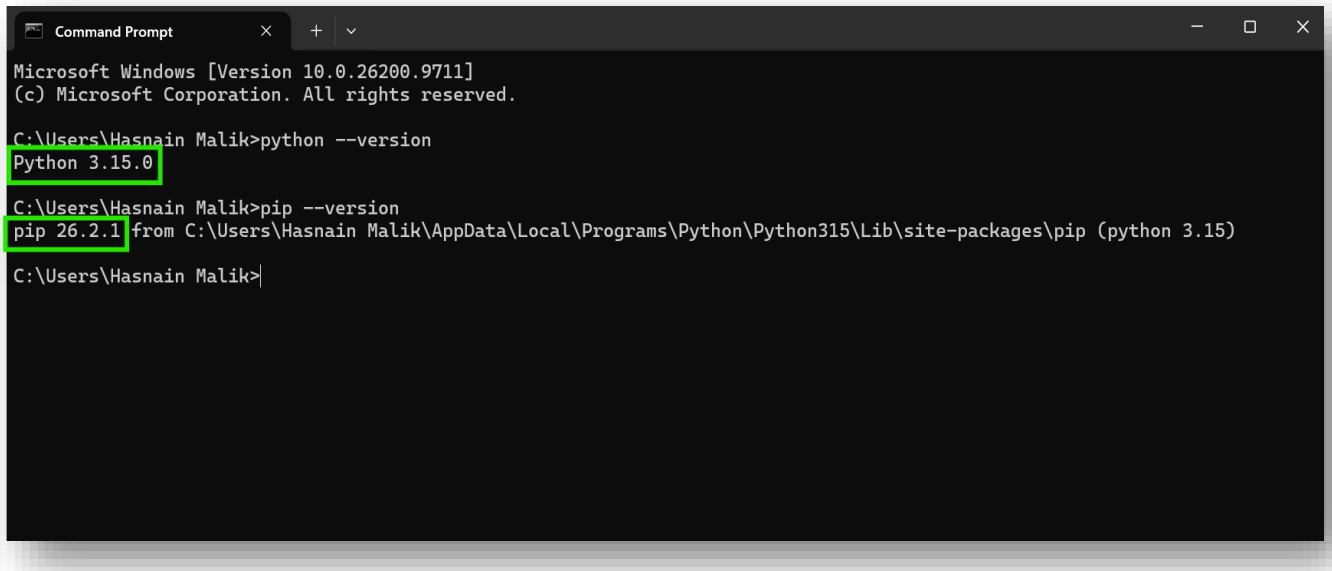


Figure 1: Command Prompt showing python --version and pip --version

Section 2: VS Code Setup & Basic Script Execution

Configured Visual Studio Code with the Python extension and selected the correct Python 3.15 interpreter. Executed a basic test script (hello.py) to confirm workspace code execution.

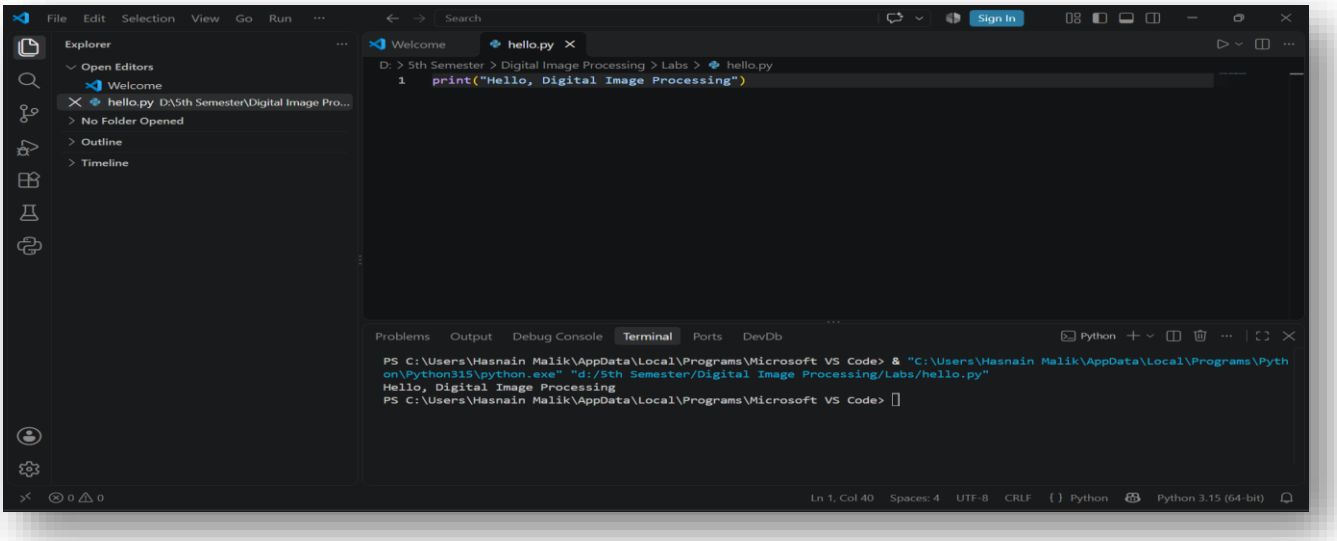


Figure 2: VS Code showing hello.py, terminal output and the active Python 3.15 interpreter

Section 3: Core Packages Installation (NumPy, Pillow, Matplotlib)

Installed core Python numerical, image manipulation, and plotting libraries :

- (NumPy, Pillow, Matplotlib) via the pip package manager.

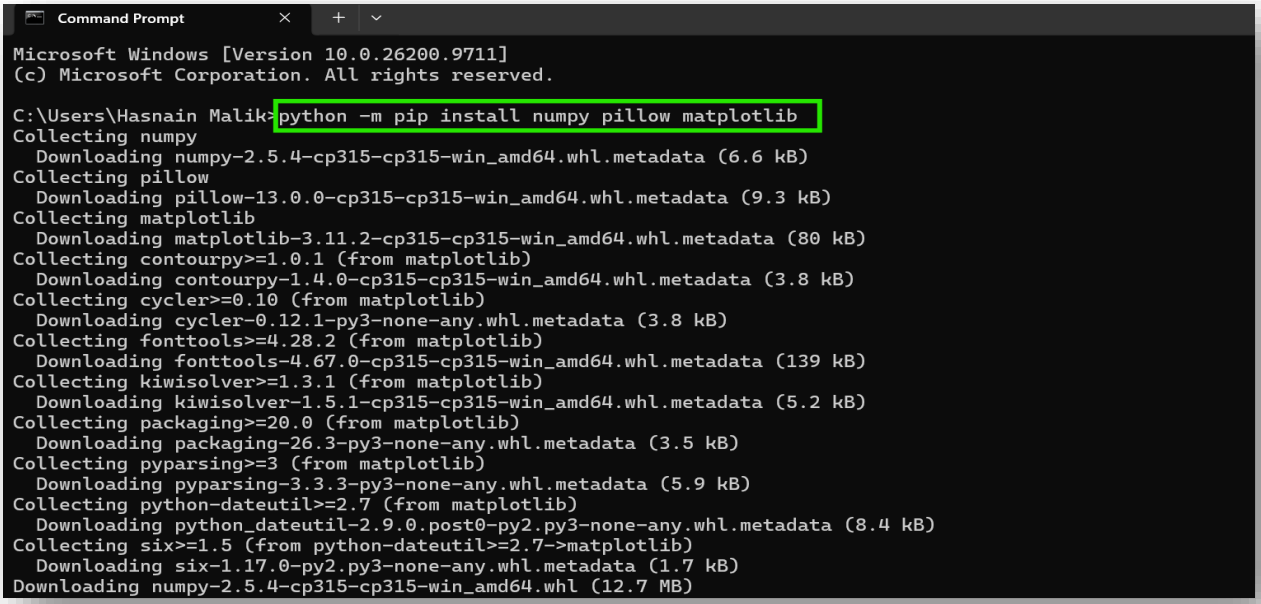


Figure 3: Installation command: python -m pip install numpy pillow matplotlib

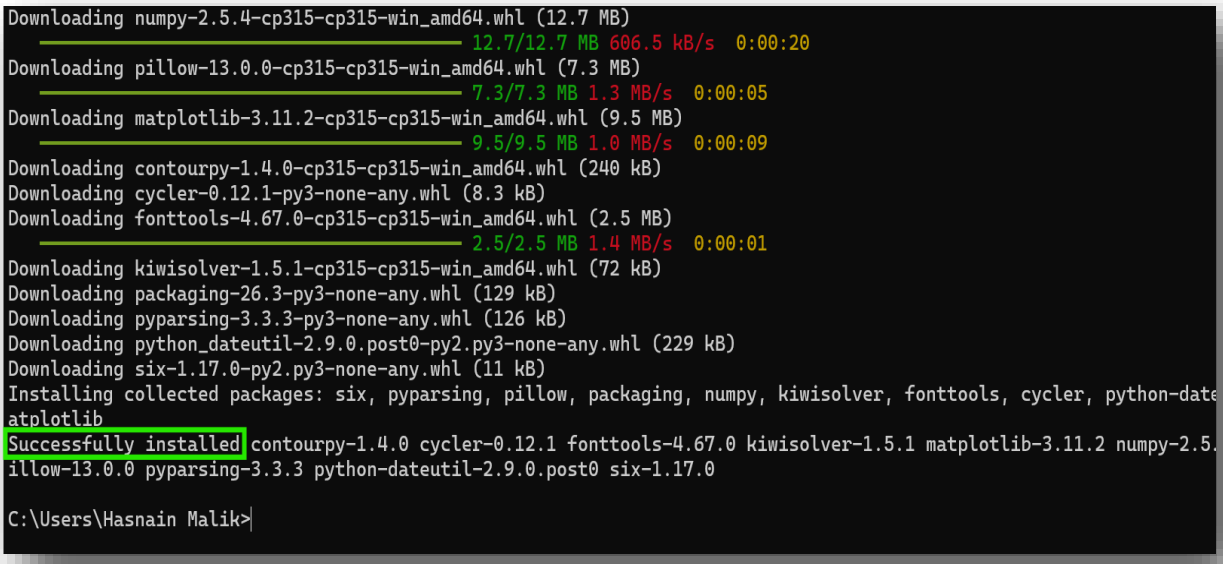


Figure 4: Successful installation confirmation message

Section 4: Library Import Validation

Created and executed lab01_imports.py containing essential import statements for:

- NumPy, Pillow, and Matplotlib to verify seamless module integration.

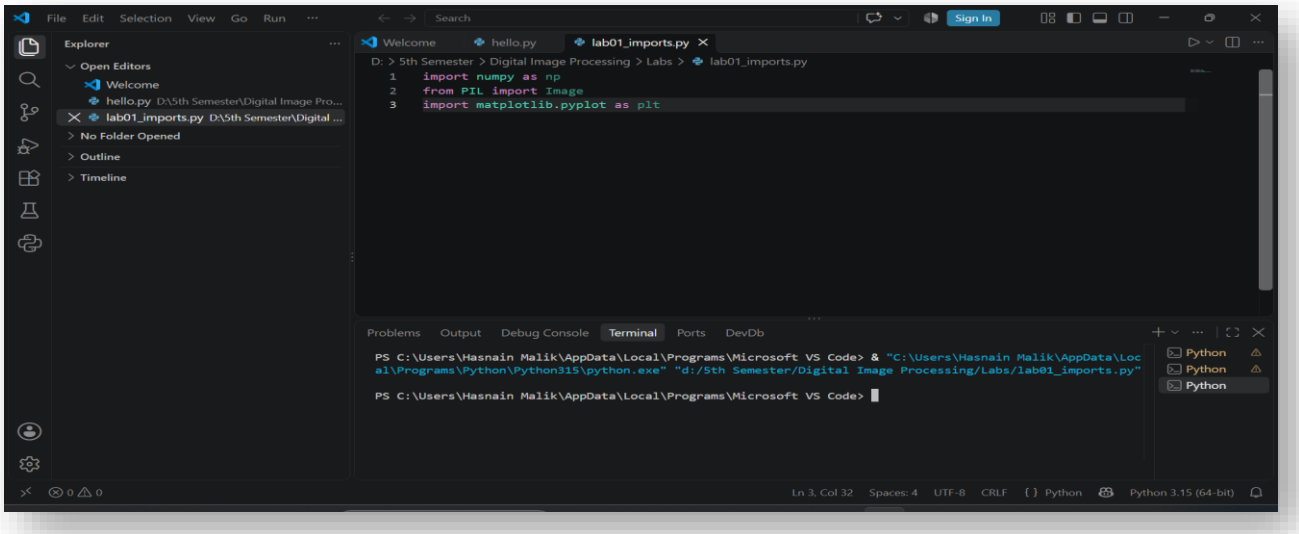


Figure 5: lab01_imports.py with import statements and error-free terminal execution

Section 5: Theoretical Review & Concepts

Q1: What is the difference between Python and pip?

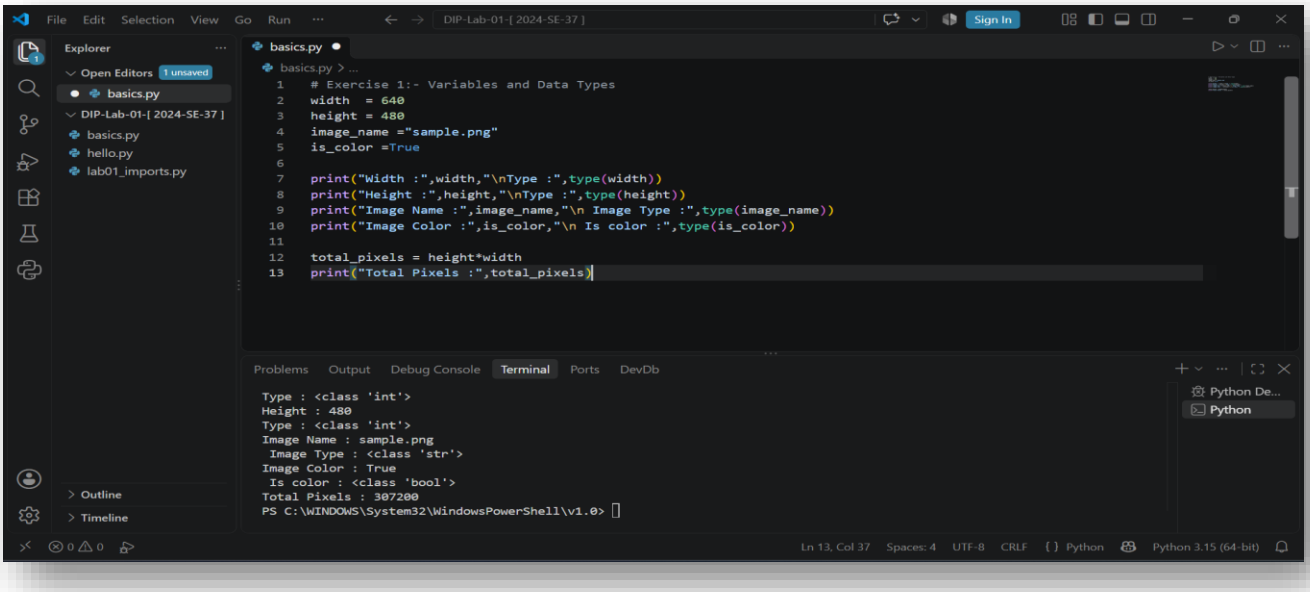
Answer: Python is a high-level programming language and runtime interpreter used to write and execute software code, whereas pip is Python's package manager utilized for downloading, installing, and managing third-party libraries and modules from PyPI.

Q2: What is the purpose of import statements in Python?

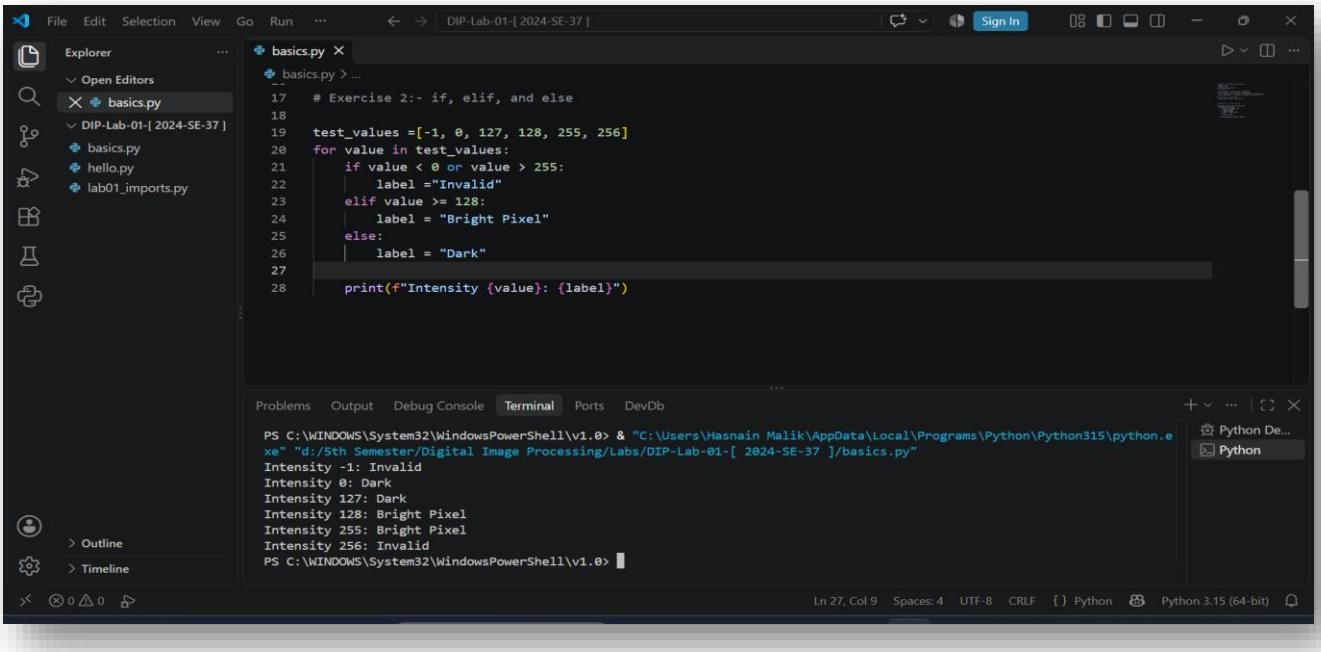
Answer: Import statements allow developers to include external libraries, modules, and pre-built packages into their current script, enabling access to specialized functions, numerical arrays, and image processing tools without rewriting code from scratch.

Section 6:- Code Exercises

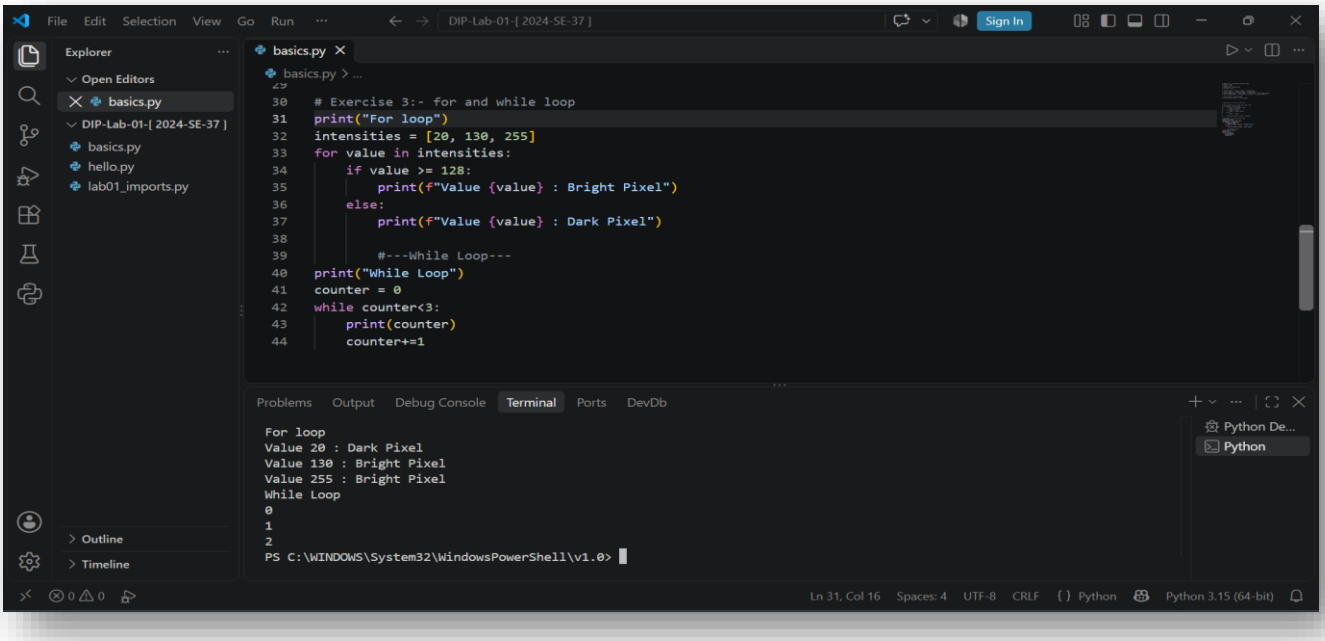
⇨ Exercise 1:



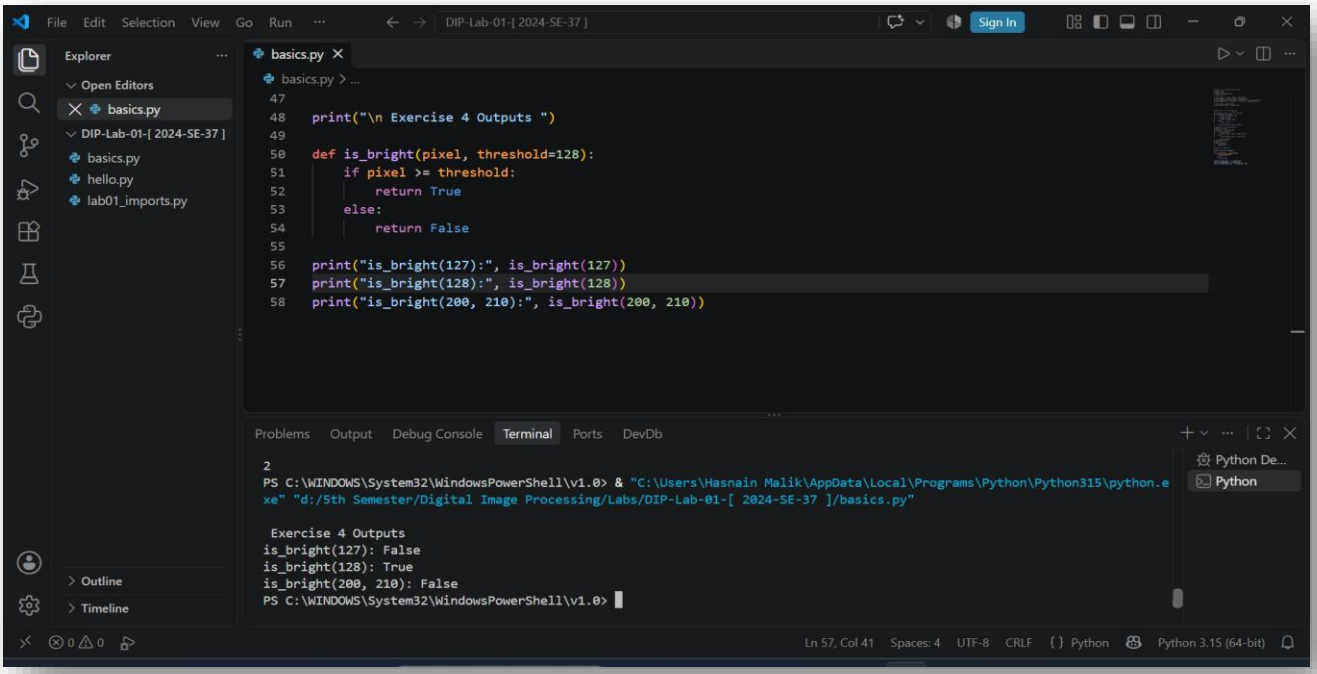
⇨ Exercise 2:



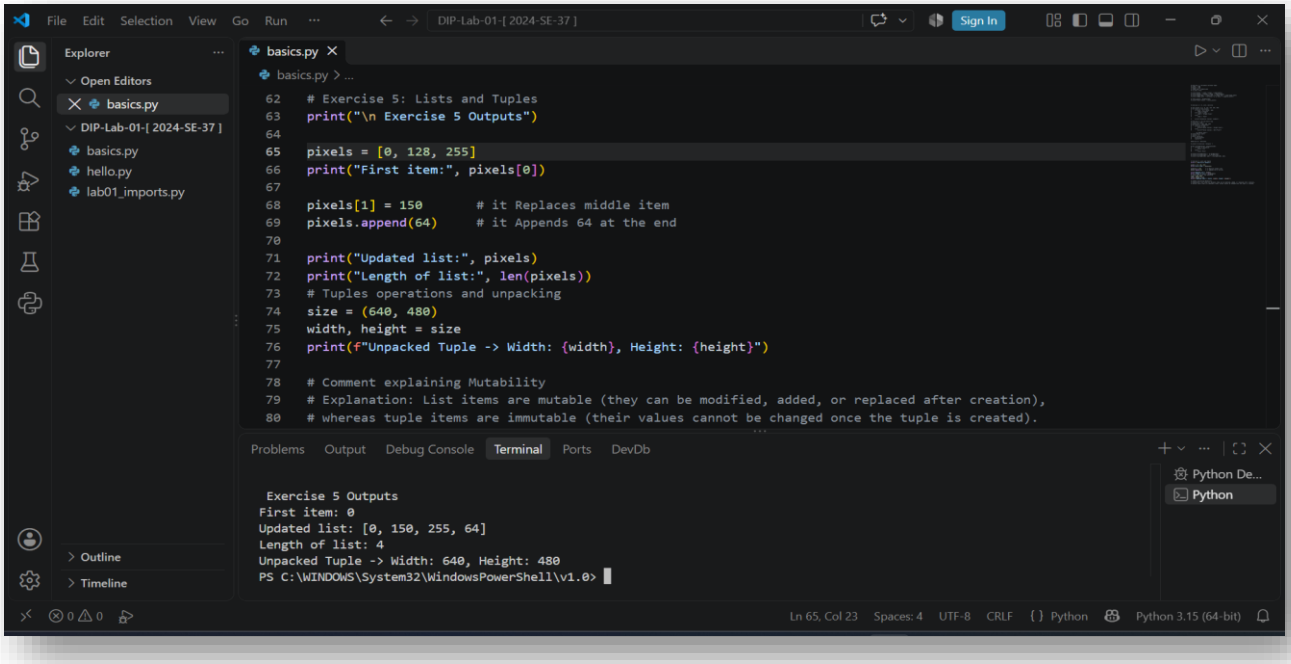
⇨ Exercise 3:



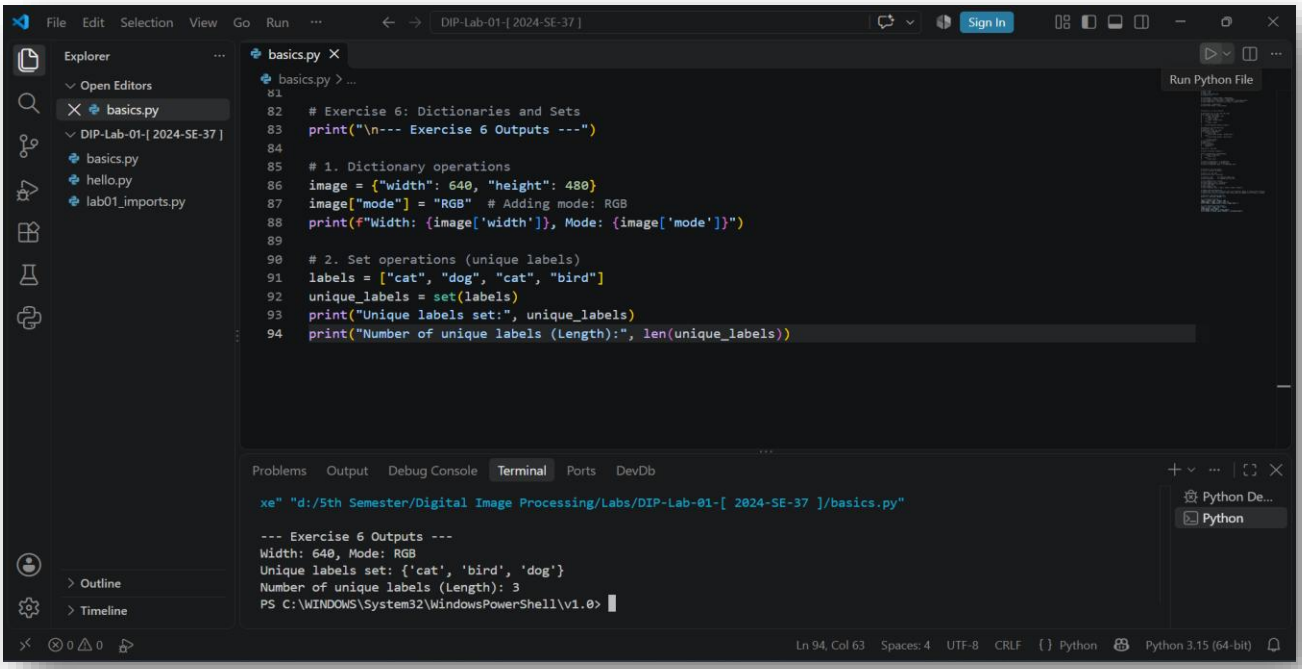
⇨ Exercise 4:


File Edit Selection View Go Run
DIP-Lab-01-[2024-SE-37]
Sign In
Explorer
Open Editors
basics.py
DIP-Lab-01-[2024-SE-37]
basics.py
hello.py
lab01_imports.py
basics.py
basics.py > ...
47
48 print("\n Exercise 4 Outputs ")
49
50 def is_bright(pixel, threshold=128):
51 if pixel >= threshold:
52 return True
53 else:
54 return False
55
56 print("is_bright(127):", is_bright(127))
57 print("is_bright(128):", is_bright(128))
58 print("is_bright(200, 210):", is_bright(200, 210))
Problems Output Debug Console Terminal Ports DevDb
2
PS C:\WINDOWS\System32\WindowsPowerShell\v1.0> & "C:\Users\Hasnain Malik\AppData\Local\Programs\Python\Python315\python.exe" "d:/5th Semester/Digital Image Processing/Labs/DIP-Lab-01-[2024-SE-37]/basics.py"
Exercise 4 Outputs
is_bright(127): False
is_bright(128): True
is_bright(200, 210): False
PS C:\WINDOWS\System32\WindowsPowerShell\v1.0>
Python De...
Python
Outline
Timeline
Ln 57, Col 41 Spaces: 4 UTF-8 CRLF { } Python Python 3.15 (64-bit)

⇨ Exercise 5:


File Edit Selection View Go Run
DIP-Lab-01-[2024-SE-37]
Sign In
Explorer
Open Editors
basics.py
DIP-Lab-01-[2024-SE-37]
basics.py
hello.py
lab01_imports.py
basics.py
basics.py > ...
62 # Exercise 5: Lists and Tuples
63 print("\n Exercise 5 Outputs")
64
65 pixels = [0, 128, 255]
66 print("First item:", pixels[0])
67
68 pixels[1] = 150 # it Replaces middle item
69 pixels.append(64) # it Appends 64 at the end
70
71 print("Updated list:", pixels)
72 print("Length of list:", len(pixels))
73 # Tuples operations and unpacking
74 size = (640, 480)
75 width, height = size
76 print(f"Unpacked Tuple -> Width: {width}, Height: {height}")
77
78 # Comment explaining Mutability
79 # Explanation: List items are mutable (they can be modified, added, or replaced after creation),
80 # whereas tuple items are immutable (their values cannot be changed once the tuple is created).
Problems Output Debug Console Terminal Ports DevDb
Exercise 5 Outputs
First item: 0
Updated list: [0, 150, 255, 64]
Length of list: 4
Unpacked Tuple -> Width: 640, Height: 480
PS C:\WINDOWS\System32\WindowsPowerShell\v1.0>
Python De...
Python
Outline
Timeline
Ln 65, Col 23 Spaces: 4 UTF-8 CRLF { } Python Python 3.15 (64-bit)

⇨ Exercise 6:


File Edit Selection View Go Run
DIP-Lab-01-[2024-SE-37]
Sign In
Explorer
Open Editors
basics.py
DIP-Lab-01-[2024-SE-37]
basics.py
hello.py
lab01_imports.py
basics.py
Run Python File
82 # Exercise 6: Dictionaries and Sets
83 print("\n--- Exercise 6 Outputs ---")
85 # 1. Dictionary operations
86 image = {"width": 640, "height": 480}
87 image["mode"] = "RGB" # Adding mode: RGB
88 print(f"Width: {image['width']}, Mode: {image['mode']}")
90 # 2. Set operations (unique labels)
91 labels = ["cat", "dog", "cat", "bird"]
92 unique_labels = set(labels)
93 print("Unique labels set:", unique_labels)
94 print("Number of unique labels (Length):", len(unique_labels))
Problems Output Debug Console Terminal Ports DevDb
xe" "d:/5th Semester/Digital Image Processing/Labs/DIP-Lab-01-[2024-SE-37]/basics.py"
--- Exercise 6 Outputs ---
Width: 640, Mode: RGB
Unique labels set: {'cat', 'bird', 'dog'}
Number of unique labels (Length): 3
PS C:\WINDOWS\System32\WindowsPowerShell\v1.0>
Outline
Timeline
Ln 94, Col 63 Spaces: 4 UTF-8 CRLF { } Python Python 3.15 (64-bit)